

SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING /
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22

Programme: B.Tech (Data Science)

Year: II Semester: IV

Subject: Database Management System

Date: 12 April 2022

Marks: 100

Time: 10.00 am to 1.00 pm

Durations: 3 (hrs)

No. of Pages: 03

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO-1 ; SO-1 ; BL-1	a.	Write down the features of Extended Entity-Relationship (EE-R) Model.	[5]
CO-4 ; SO-1 ; BL-2	b.	Define indexing and explain ordered indices in detail.	[5]
CO-1 ; SO-1 ; BL-1	c.	Describe the three-schema architecture. Why do we need mappings between schema levels?	[5]
CO-2 ; SO-2 ; BL-2	d.	Explain in detail about various key constraints used in the database system.	[5]
Q2 CO-1; SO-1; BL-1	a.	List down advantages and disadvantages of DBMS. Discuss any five advantages by comparing with file systems.	[10]

CO-1; SO-1; BL-2	b.	Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular period of time, and has an associated due date, and the date when the payment was received.	[10]
Q3 CO-3; SO-2; BL-3	a.	What is the need for normalization? Explain the first, second and third normal forms with examples.	[10]
CO-3; SO-1; BL-3	b.	Consider a relation R(A,B,C,D,E) with FDs {AB → C, DE → C, and B → D} i. Find the candidate keys of the above relation. ii. Indicate all BCNF violations for R. iii. Decompose the relations into collections of relations that are in BCNF. iv. Indicate which dependencies if any are not preserved by the BCNF decomposition.	[10]
Q4 CO-2; SO-2; BL-3	a.	Consider the following six relations for an order processing database application in a company: Customer (<u>Cust#</u> , Cname, City) Order(<u>Order#</u> , Odate, Cust#, Ord_Amt) Order_Item(Order#, <u>Item#</u> , Qty) Item(<u>Item#</u> , Unit_Price) Shipment(<u>Order#</u> , <u>Warehouse#</u> , Ship_date) Warehouse(<u>Warehouse#</u> , City) Ord_Amt refers to the total dollar amount of an order; Odate is the date the order was placed; Ship_date is the date an order is shipped from the warehouse. An order can be shipped from any warehouse. Specify the following queries in SQL:	[15]

		i. Write down the primary key and foreign key of all tables in the database. ii. List the Order# and Ship_date for all orders shipped from Warehouse# "W2". iii. List the Warehouse information from which the Customer named "Jose Lopez" was supplied his orders. Produce a listing of Order#, Warehouse#. iv. List the Order# for orders that were shipped from warehouses in New York. v. List all items that have a price greater than the average price. vi. Find the item with the maximum unit price. vii. List all customer names whose orders were shipped from a warehouse in the same city as they live in.	
CO2-; SO-2; BL-3	b.	List the aggregate functions supported by SQL?	[5]
Q5 CO2-; SO-2; BL-3	a.	Explain the following commands with example in SQL. i. Drop command ii. Delete command iii. Truncate Command iv. Update v. Alter	[10]
CO-4; SO-1; BL-2	b.	Why do we need concurrency control? Illustrate Concurrent execution of transactions with examples?	[10]
Q6 CO-4; SO-1; BL-2	a.	Define transaction and discuss the states of transaction with suitable diagrams. Explain ACID properties of database transactions.	[10]
CO-4; SO-1; BL-2	b.	Briefly explain the two phase locking protocol and strict two phase locking protocols used in concurrency control?	[10]